Agentic AI Final Project: Proposal – Hrithik Kathiresan

1. Problem Statement & Motivation

Like me, there are many college students who struggle to cook nutritionally balanced, varied, and tasty meals every day. Limited time, poor cooking skills, budget constraints, and a lack of options make it difficult to cook the right meals. Because of this, many students end up eating out or ordering takeout often, which can be expensive and unhealthy.

Guides, recipe websites, and apps exist, but they typically assume that people already know what they want to cook or have a fridge stocked full of ingredients. They don't adapt to a college student's unpredictable collection of food.

The primary target users are college students living in dorms or apartments who have limited cooking experience, only basic kitchen tools, inconsistent ingredients, and want quick and practical meals based on what they already have.

This problem is very well suited for an agentic AI system because the workflow is dynamic, it is multi-step, and very context dependent. Instead of following a fixed, predefined flow, the system must:

- Interpret user limitations (ingredients, skill level, time, tools, etc.)
- Plan possible meals and evaluable feasibility and difficulty
- Interact with tools (image analysis and ingredient tracking)
- Adapt recommendations based on user input changes

A traditional rule-based system or static recipe app cannot flexibly reason across these constraints. An agentic AI system with specialized agents can autonomously coordinate perception, planning, evaluation, and instruction generation, making it more effective and scalable than a manually defined workflow.

2. Use Cases & Scenarios

Scenario 1: Ingredient-Based Meal Planning

A college student wants to cook dinner using only the limited number of ingredients they currently have. The student enters the list of the ingredients they have, their skill level, their tools, and the time they have to cook.

Next, the agent:

- Interprets the constraints and ingredient availability
- Generate multiple feasible meal options
- Rank meals based on simplicity, time, and skill level
- Present suggestions to the user
- Provide step-by-step cooking instruction once a meal is selected

Scenario 2: Fridge Photo Ingredient Detection

A student wants quick meal ideas without manually listing out each ingredient in their fridge or cupboard. The student uploads a photo of their fridge and cupboard.

Next, the agent:

- Analyzes the image to detect visible ingredients
- Converts detected items into a list of ingredients
- Asks user for clarification if uncertain about items
- Proceed with meal planning using ingredient list

- Recommend meals based on cooking constraints

3. Initial System Design

Preliminary Agents:

1. User Interaction Agent – Handles user input, asks clarifying questions, and manages conversation flow
2. Ingredient Management Agent – Maintains and updates ingredient inventory
3. Vision/Ingredient Detection Agent – Processes fridge images
4. Meal Planning Agent – Generates feasible meal options and ranks options based on skill level and time
5. Instruction Generation Agent – Produces detailed cooking instructions
6. Critic/Validator Agent – Reviews generated recipes for feasibility. Also checks for missing steps, unrealistic assumptions, or unsafe cooking advice

Current tools, APIs, and data sources: Vision model API for ingredient recognition from images web search for recipes

4. Feasibility & Risks

Potential Challenges – Vision model may misidentify items in fridge photos and some generated recipes may assume ingredients or tools the user does not have

Constraints – Rate limits on vision or LLM APIs, privacy concerns for user-uploaded images, computation constraints

Timeline – Following suggested course guidelines for this project